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1000MCN401122201

Pages: 2

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree (S, FE) Examination May 2023 (2019 Scheme)

Course Code: MCN401

Course Name: INDUSTRIAL SAFETY ENGINEERING

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|--|-----|
| 1 | Write the importance of safety in organisations | (3) |
| 2 | How can you describe safety policy | (3) |
| 3 | How do you monitor the safety performance in industries | (3) |
| 4 | Write the significance of work permit system | (3) |
| 5 | How can workers' safety be ensured during underpinning work? | (3) |
| 6 | List the various hazards of underwater works | (3) |
| 7 | What are the various objectives of Maintenance | (3) |
| 8 | List the safety precautions to be followed during grinding operation | (3) |
| 9 | What do you meant by Hazard and Risk | (3) |
| 10 | What is meant by MSDS | (3) |

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

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|----|---|-----|
| 11 | a) What are the causes of industrial accidents | (7) |
| | b) Explain the various theories of accident causation | (7) |

OR

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|----|---|------|
| 12 | With the help of a neat sketch explain safety organization structure. Also write the importance of safety organization structure. | (14) |
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Module II

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| 13 | Classify personal protective equipment used in industries. List the suitability of at least seven types of PPEs. | (14) |
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OR

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| 14 | a) Explain the role of management & employees in housekeeping? | (7) |
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- b) What procedures should be taken during confined space access to protect a worker's life? (7)

Module III

- 15 a) Identify various hazards that may arise during the various stages of building construction. (7)
- b) Mention the safety precautions that will be implemented to avoid scaffolding-related incidents. (7)

OR

- 16 Discuss the important types of ergonomic hazards associated with industries. (14)
- How can we reduce its impact?

Module IV

- 17 Explain the safety precautions to be followed while doing various material handling assessments and techniques in industries (14)

OR

- 18 a) Briefly explain the maintenance of chains slings (7)
- b) Briefly explain the maintenance of clamps (7)

Module V

- 19 a) What is meant by HAZOP? How do you conduct a HAZOP analysis? (7)
- b) Briefly explain Criticality Analysis (7)

OR

- 20 How do you classify fires and explain various types of fire extinguishers used in industries? (14)
